

Final Project Report

Global Malnutrition Trends: A Power BI Analysis (1983-2019)

1. Introduction

1.1 Project Overview

Malnutrition remains a pressing global issue affecting millions of children under the age of five. It leads to long-term health, cognitive, and economic consequences for individuals and communities. This Power BI project aims to visualize malnutrition data collected from 1983 to 2019 across various countries, indicators, and populations. The goal is to identify countries at highest risk, analyze key patterns, and support better policy and program decisions.

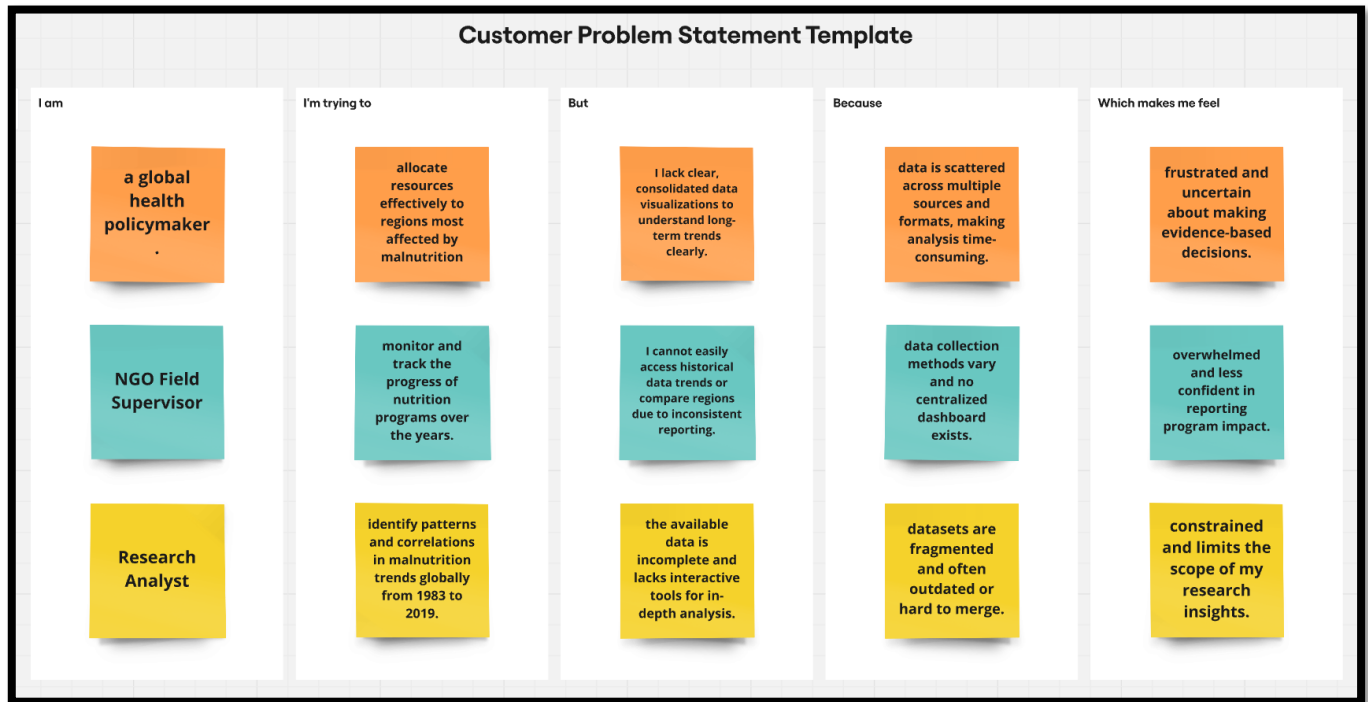
1.2 Project Objectives

- To explore the trends of major malnutrition indicators: stunting, wasting, underweight, overweight, and severe wasting.
- To identify the top and bottom countries based on these indicators.
- To build an interactive Power BI dashboard for effective storytelling and visual understanding.
- To generate actionable insights through visual reports and summaries.
- To improve awareness around global disparities in nutrition levels using real data.

2. Project Initialization and Planning Phase-

2.1 Define Problem Statements:

Global health policymakers and NGOs face significant challenges in understanding malnutrition trends over time due to fragmented and inconsistent data sources. This lack of accessible, comprehensive, and clear visualizations makes it difficult to identify high-risk regions and allocate resources effectively. As a result, they feel frustrated and uncertain about making informed decisions to combat malnutrition globally.



2.2 Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To analyse global malnutrition trends from 1983 to 2019 using Power BI, identify patterns and disparities across regions, and provide visual insights for policy-making and awareness.
Scope	This project will analyze malnutrition data from the Kaggle Dataset [World Bank, WHO, and UNICEF (1983-2019)]. It will provide visualizations of prevalence rates, regional disparities, and correlations with socioeconomic indicators. Limitations include potential data gaps and a focus on available indicators, excluding factors not directly measured by these sources.

Problem Statement	
Description	Global malnutrition remains a persistent issue, with fragmented data sources hindering effective analysis of trends from 1983-2019. Policymakers and NGOs lack clear, consolidated visualizations to inform resource allocation and intervention strategies.
Impact	A detailed analysis can assist governments, NGOs, and researchers in identifying high-risk areas, understanding long-term patterns, and developing focused interventions. Insightful visualizations also promote public awareness.
Proposed Solution	
Approach	Collect and clean malnutrition-related data (1983–2019) from Kaggle datasets. Load data into Power BI for processing and modelling. Create interactive dashboards showcasing trends by year, region, and malnutrition type. Use filters and visuals like line charts, bar graphs, heat maps, and slicers for deeper insights.
Key Features	- Interactive dashboards displaying malnutrition prevalence rates by region and year. - Customizable visualizations allowing users to explore correlations with socioeconomic indicators. - Trend analysis tools for identifying malnutrition hotspots and tracking progress over time.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Standard desktop computer	11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	477 GB

Software		
Frameworks	N/A	-
Libraries	DAX	Built into Power BI
Development Environment	BI Tool	Power BI Desktop
Data		
Data	Source	e.g., Kaggle dataset

2.3 Initial Project Planning:

Sprint (Phase)	Epic	User Story Number	Task / User Story	Story Points	Priority	Assigned To	Start Date	End Date
Sprint 1	Project Initialization and Planning	USN-1	Set up project workspace and tools, ensuring all team members have access.	2	High	Ratnadeep, Mital	24-04-2025	24-04-2025
		USN-2	Draft and finalize the problem statement and project objectives with team input.	3	High	Ratnadeep, Mital	24-04-2025	25-04-2025
		USN-3	Assign roles, responsibilities, and define communication channels for the team.	2	Medium	Ratnadeep	24-04-2025	25-04-2025
		USN-4	Prepare a detailed project proposal and get stakeholder approval.	3	High	Ratnadeep, Mital	25-04-2025	25-04-2025
Sprint 2	Data Collection and Preprocessing	USN-5	Identify, gather, and document relevant malnutrition datasets from trusted global sources.	3	High	Mital, Samrat	25-04-2025	26-04-2025
		USN-6	Import raw data into Power BI and organize it for analysis.	3	High	Mital, Chaitanya	26-04-2025	27-04-2025
		USN-7	Clean and preprocess the data: handle missing values, remove duplicates, and standardize formats.	5	High	Samrat, Chaitanya	27-04-2025	28-04-2025
		USN-8	Document data sources, preprocessing steps, and any assumptions made during cleaning.	2	Medium	Mital	28-04-2025	28-04-2025

Sprint 3	Data Visualization	USN-9	Design initial charts and graphs to represent malnutrition trends by year, region, and demographic.	4	High	Ratnadeep, Mital	28-04-2025	29-04-2025
		USN-10	Develop interactive visuals (filters, slicers, maps) for deeper data exploration in Power BI.	4	High	Mital, Samrat	29-04-2025	29-04-2025
		USN-11	Review and refine visualizations based on team feedback.	2	Medium	Ratnadeep, Chaitanya	29-04-2025	29-04-2025
Sprint 4	Dashboard	USN-12	Create a comprehensive dashboard integrating all key visuals and KPIs.	5	High	Ratnadeep, Mital	29-04-2025	30-04-2025
		USN-13	Ensure dashboard interactivity and user-friendliness for stakeholders.	3	High	Mital, Chaitanya	30-04-2025	30-04-2025
		USN-14	Test dashboard performance and make adjustments as needed.	2	Medium	Samrat, Chaitanya	30-04-2025	30-04-2025
Sprint 5	Report	USN-15	Draft a detailed report summarizing findings, insights, and recommendations from the analysis.	4	High	Ratnadeep, Mital	30-04-2025	30-04-2025
		USN-16	Review and finalize the report with input from all team members.	2	High	All	30-04-2025	30-04-2025
		USN-17	Prepare a presentation or summary slides for stakeholders.	2	Medium	Ratnadeep, Mital	30-04-2025	30-04-2025

3. Data Collection and Preprocessing Phase:

3.1 Data Collection Plan & Raw Data Sources Identification Template:

• Data Collection Plan Template

Section	Description
Project Overview	This project aims to analyze global malnutrition trends from 1983 to 2019 using data analytics and visualization tools. The objective is to uncover patterns and insights to support policy making and awareness.
Data Collection Plan	Data will be collected from the publicly available Kaggle dataset "Malnutrition Across the Globe," which compiles malnutrition statistics for multiple countries and years.
Raw Data Sources Identified	Kaggle Dataset: "Malnutrition Across the Globe" (https://www.kaggle.com/datasets/ruchi798/malnutrition-across-the-globe), containing country-wise and year-wise malnutrition indicators in CSV format.

• Raw Data Sources Template

Source Name	Description	Location /URL	Format	Size	Access Permissions
Kaggle - Malnutrition Across the Globe	Contains global malnutrition statistics (stunting, wasting, underweight) for multiple countries and years (1983-2019). Suitable for direct import and analysis in Power BI.	Link	CSV	~1 MB	Public (Kaggle login required)
...	...	...	...	...	...

3.2 Data Quality Report Template

Data Source	Data Quality Issue	Severity	Resolution Plan
Kaggle Dataset	Missing values in prevalence rates for certain countries and years; inconsistencies in country names; potential outliers.	Low/ Moderate/ High	1. Missing Values (Easy): Use the "Replace Values" feature in Power BI's Power Query Editor to replace blank values with the average of the previous and next available years for that country or region. 2. Country Name Inconsistencies (Easy): In Power Query Editor, use the "Replace Values" feature to standardize country names (e.g., replace "USA" with "United States"). Create a separate table with correct ISO codes and merge it using country names to get a standardized column. 3. Outliers (Mod): Create box plots in Power BI visualizations to identify outliers. Investigate these points, and if validated as errors, use the "Replace Values" feature in Power Query Editor to replace them with corrected data or a calculated average.

3.3 Data Exploration and Preprocessing Template

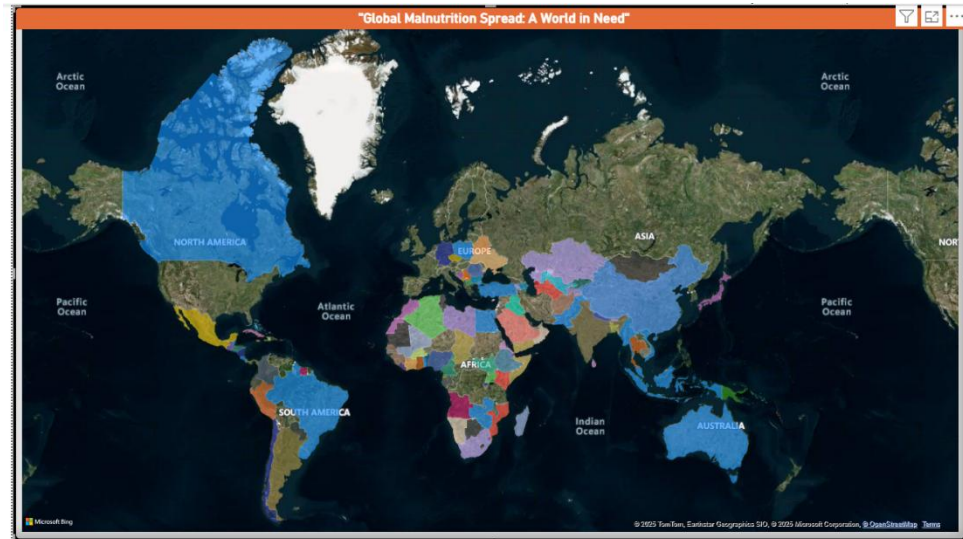
Section	Description
Data Overview	The dataset "Malnutrition Across the Globe" from Kaggle provides malnutrition statistics (stunting, wasting, underweight) for various countries and years (1983-2019). We will use it to analyze global malnutrition trends using Power BI.
Data Cleaning	Handle missing values by imputing using forward or backward fill from adjacent years. Identify and remove or correct outliers based on cross-validation with WHO/UNICEF data. Standardize inconsistent country names using a mapping table with ISO codes.

Data Transformation	Use Power Query Editor to filter data by year and region. Pivot data to create summary tables and unpivot it back for charting. Create calculated columns for new metrics (e.g., malnutrition prevalence change over time). Sort to highlight important data.
Data Type Conversion	Ensure all columns have the correct data types (e.g., numeric columns are set as decimal numbers, date columns are set as dates). Correct data types by using Power Query Editor for data validation and Power BI's transformation tools for accurate data handling
Column Splitting and Merging	No column splitting or merging is anticipated at this stage. If needed, create calculated columns for merged data (e.g., combining regions). Use split to split data into multiple columns.
Data Modeling	Define relationships between the country table, year table, and malnutrition data table. Create measures using DAX (e.g., average malnutrition rate, year-over-year change) to enable advanced analysis and create interactive visualizations.
Save Processed Data	Save the cleaned and processed data as a Power BI Desktop file (.pbix). Consider exporting cleaned data as a CSV file or Power BI dataflow for future use in other projects.

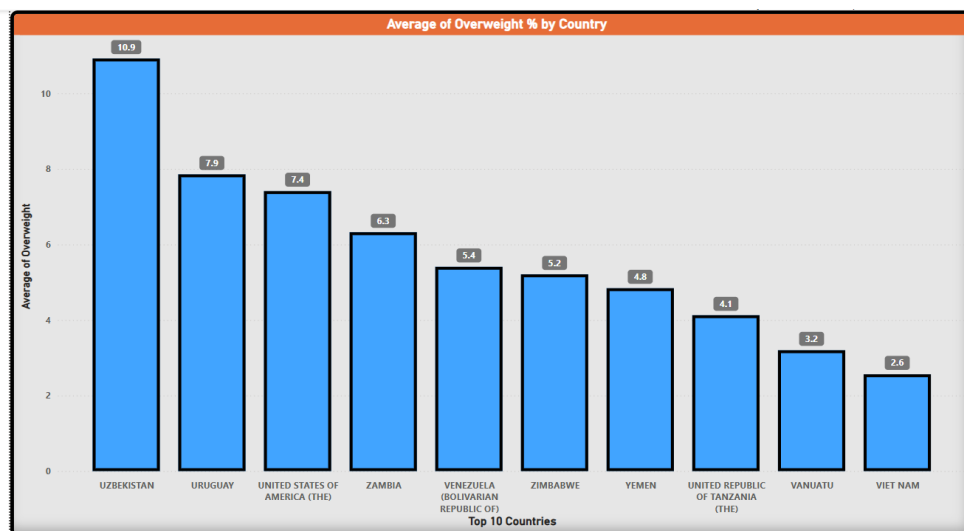
4. Data Visualization-

Business Questions and Visualisation

- Which countries have the highest rates of malnutrition (wasting)?
 - **Visualization:** Filled Map Showing “Global Malnutrition Spread: A World in Need”

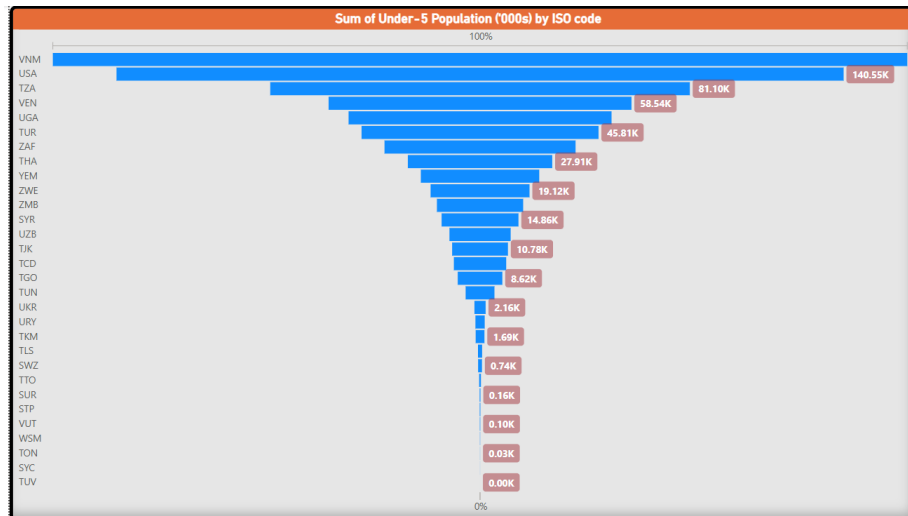


- What is the Average Overweight % in each Country?
 - **Visualization:** clustered Column Chart Showing Average of Overweight



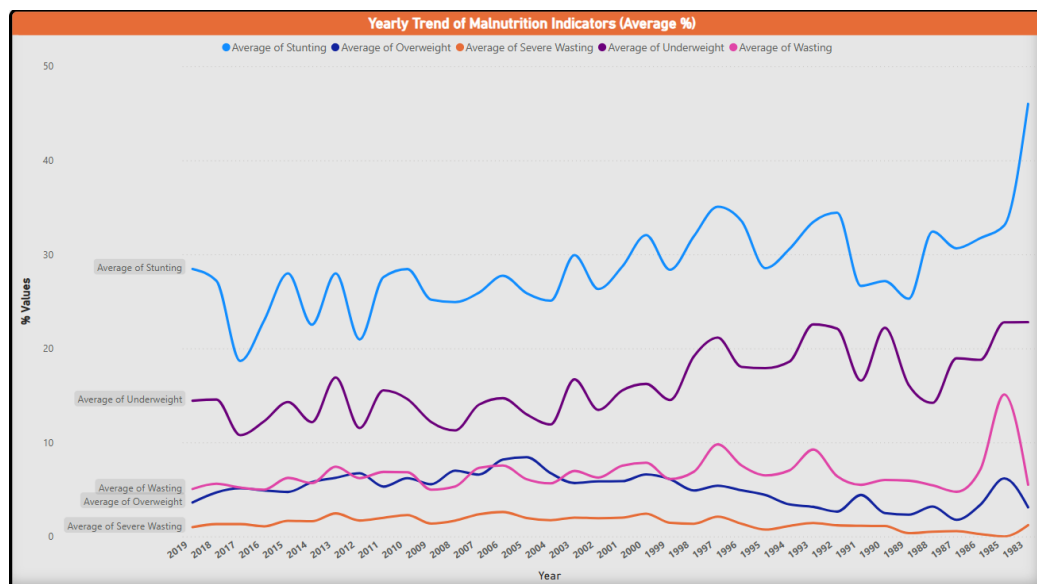
- How Does the Population of children under 5 vary across different countries (represented by ISO Code)?

Visualization: Funnel Chart Showing Sum of under 5 Population



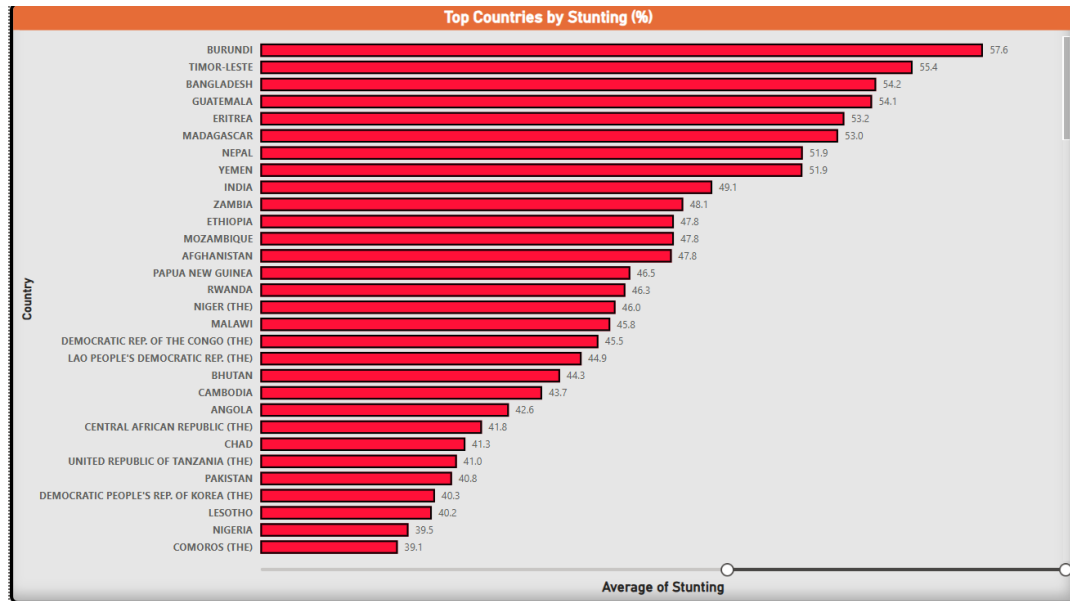
- How has malnutrition changed year by year?

Visualization: Line Chart Showing Year trend



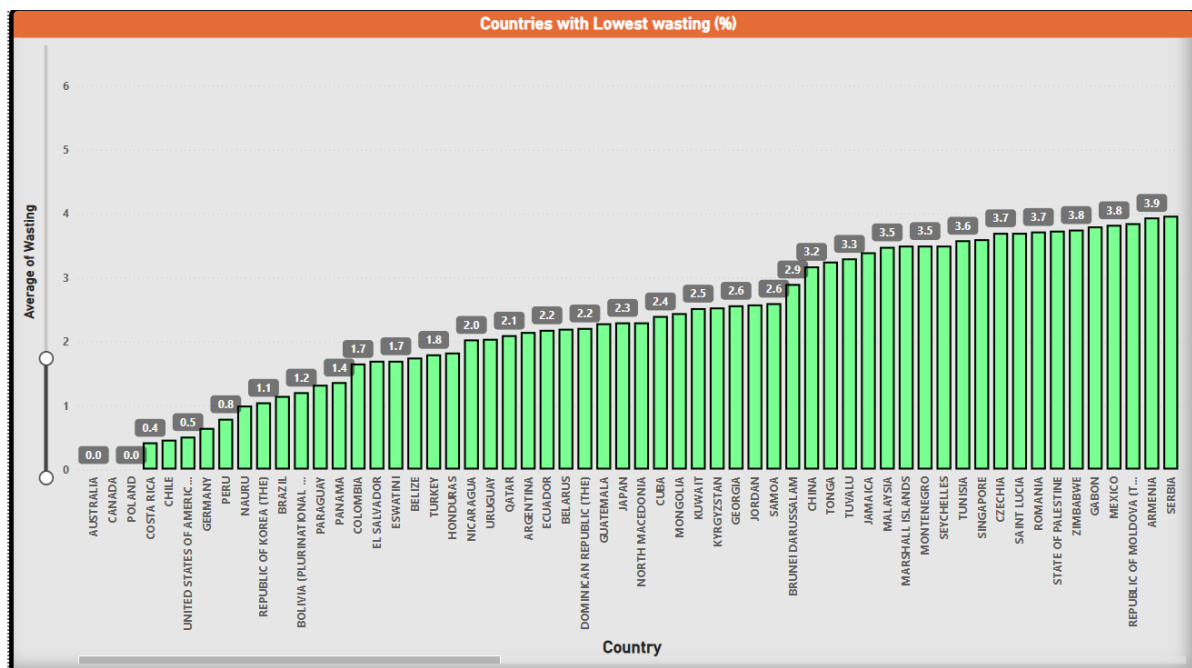
- Which Countries have the most stunting problems?

Visualization: Horizontal bar chart Showing Stunting problem

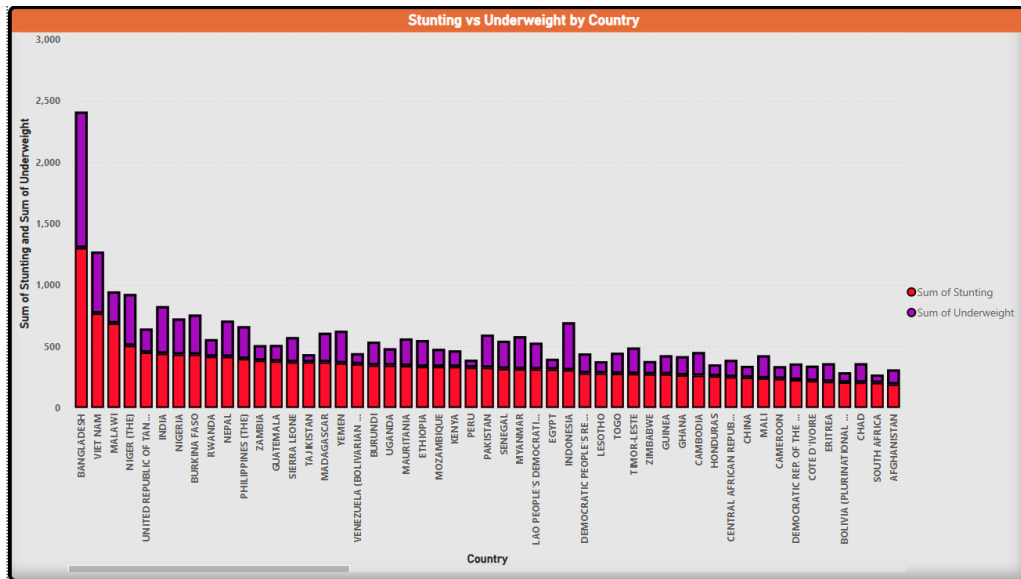


- Which countries have the least wasting problems?

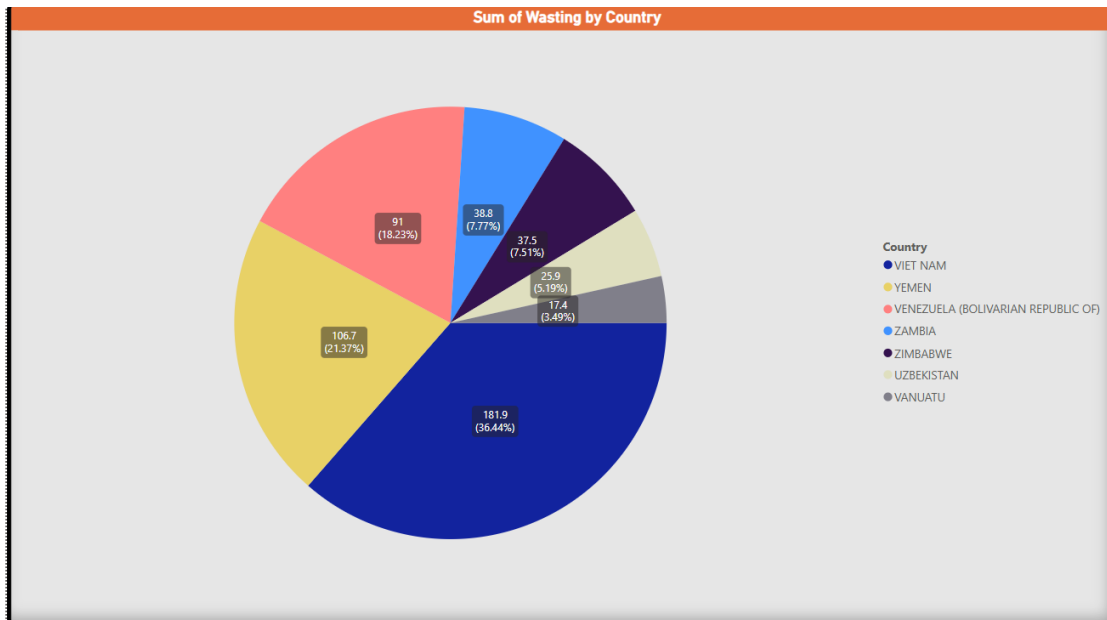
Visualization: Vertical bar chart showing least wasting problem



- Visualization:** Stacked Column Chart showing child stunting vs underweight

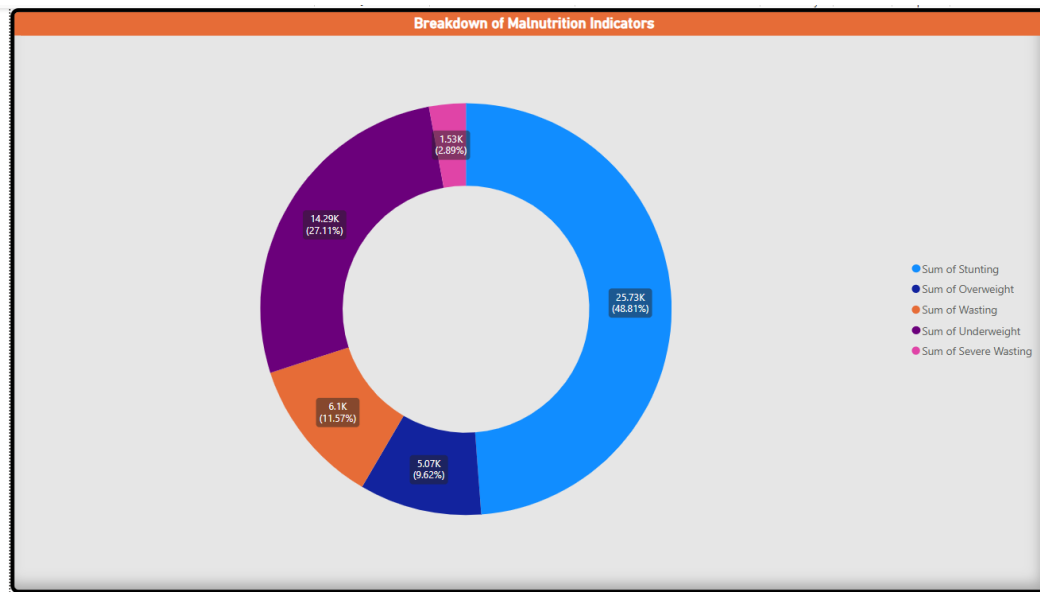


- Visualization:** Pie Chart Showing sum of wasting by country



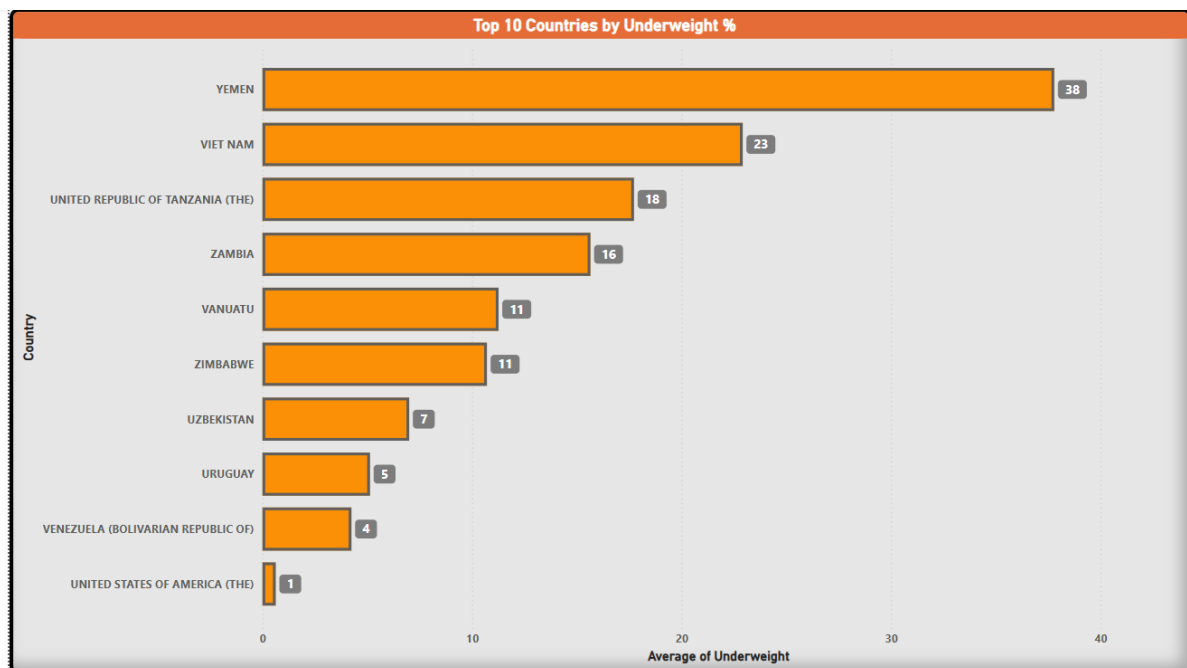
- Which Specific indicator requires the most urgent attention in different countries?

Visualization: Donut chart showing Breakdown of malnutrition indicators



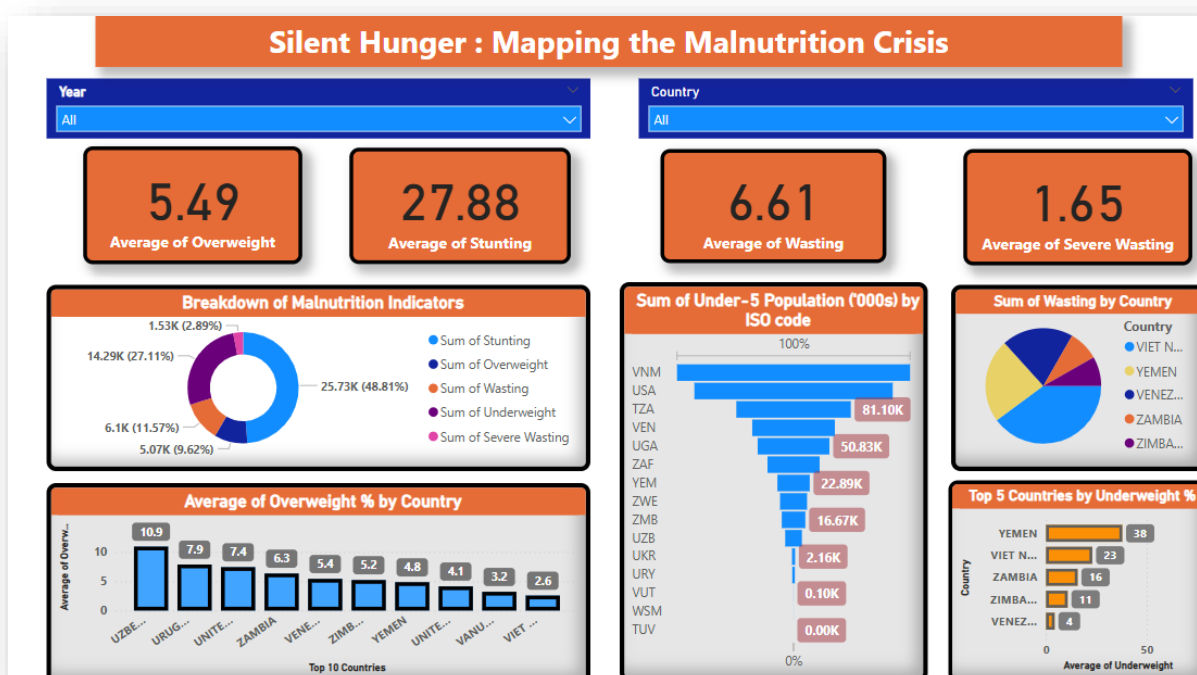
- Which 10 countries have the highest underweight percentage?

Visualization: Clustered bar Chart showing the top 10 countries



5. Dashboard:

5.1. Dashboard Design File-



Here are five potential outcomes from the dashboard image provided:

1. Yemen Has the Highest Underweight Percentage

- The **Top 5 Countries by Underweight %** chart highlights Yemen with the **highest value (38%)**.
- This signals **severe nutritional challenges** and makes Yemen a critical focus for intervention.

2. Vietnam Has the Largest Under-5 Population in the Dataset

- From the **bar chart on Under-5 Population by ISO code**, **Vietnam (VNM)** tops the chart.
- This suggests that **policy decisions and support** in Vietnam can have a high impact on global child health numbers.

3. Stunting is the Most Prevalent Malnutrition Indicator

- The **Donut Chart Breakdown** shows **stunting** occupies **nearly half the total cases (48.81%)**.
- This indicates that **chronic malnutrition** is a **long-term global issue** that demands sustained efforts.

4. Overweight Percentages Are Rising in Some Countries

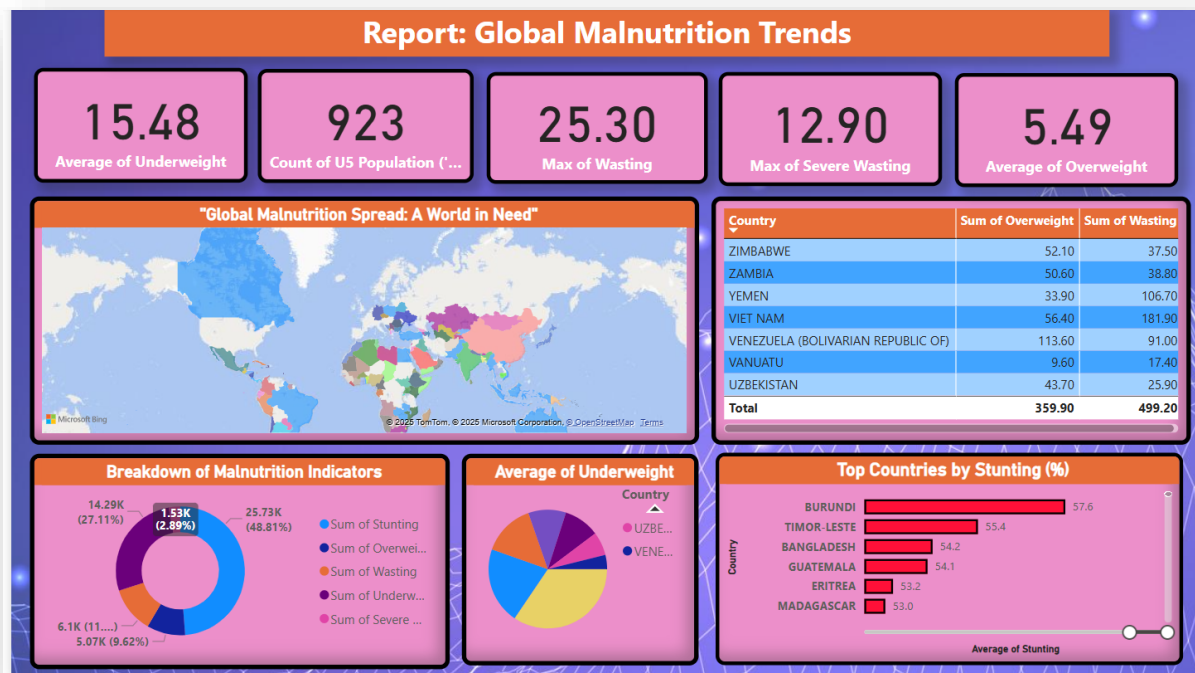
- The **Top Countries by Overweight %** chart reveals that countries like **Uzbekistan, Uruguay, and the USA** have relatively high overweight values.
- This reflects the **dual burden of malnutrition** — both undernutrition and overnutrition exist in some regions.

5. Wasting Remains an Issue in Specific Countries

- The **Pie Chart on Wasting by Country** shows countries like **Vietnam, Yemen, and Venezuela** with high shares of wasting.
- Wasting is often linked to **acute hunger** and emergency food insecurity — calling for immediate attention.

6. Report;

6.1 Report Design File-



Observations from Power BI Report – “Global Malnutrition Trends”

1. Global Overview of Key Indicators

The dashboard highlights average malnutrition indicators globally:

- **Average Underweight:** 15.48%
- **Average Overweight:** 5.49%
- **Maximum Wasting:** 25.30%
- **Maximum Severe Wasting:** 12.90%
- The data also includes a **total under-5 population count** of 923, indicating the scale of data coverage.

These metrics provide an at-a-glance understanding of malnutrition severity and prevalence across multiple dimensions.

2. Geographical Disparities

The **filled map visual** titled “*Global Malnutrition Spread: A World in Need*” effectively shows malnutrition distributed globally.

Countries across **Africa, South Asia, and parts of Latin America** show higher rates of malnutrition indicators, while **developed nations** show lower prevalence but rising overweight concerns.

3. High-Risk Countries Identified

The **table visualization** highlights countries with both high **overweight** and **wasting** percentages:

- **Vietnam and Venezuela** show critical values across both extremes.
- **Zambia and Yemen** also reflect dangerously high rates of undernutrition.

These countries may require **immediate intervention**.

4. Distribution of Malnutrition Indicators

The **donut chart** illustrates the proportion of each malnutrition indicator:

- **Stunting dominates** with nearly **49%** of total burden.
- Underweight and wasting are also substantial.
- Severe wasting and overweight represent a smaller share but are rising.

This breakdown emphasizes the need to **prioritize stunting-focused programs**.

5. Country-Level Breakdown

Visuals such as:

- **Top Countries by Stunting %**
- **Average of Underweight by Country**

...highlight **Burundi, Timor-Leste, and Bangladesh** as the most affected by stunting.

Countries like **Uzbekistan and Venezuela** show rising overweight percentages — a sign of the **dual burden of malnutrition**.

Conclusion

This report offers a clear, visual, and data-driven understanding of the global malnutrition crisis. It helps identify high-burden regions, track indicator trends, and guide policy prioritization. Power BI's interactive visuals make it ideal for ongoing monitoring and stakeholder engagement.

7. Performance Testing

7.1 Utilization of Data Filters

In our Power BI dashboard, we used several filters to allow users to interact with the data and narrow down specific trends.

Filters and Slicers used:

-  **Country Slicer** — allows users to select one or multiple countries to view their indicators.
-  **Year Slicer** — filters visuals to show trends for a selected year or time range.
-  **Top N Filters** — used in visuals to show the **Top 10 countries** for highest or lowest malnutrition rates (e.g., Top 10 by stunting).
-  **Indicator-Based Filters** — some visuals were built using filtered fields (e.g., only stunting or only overweight).

These filters made the dashboard more dynamic, personalized, and visually manageable.

7.2 Number of Calculated Fields (DAX Measures)

We used several **calculated fields (measures)** to simplify data and create KPIs:

Measure Name

Average Overweight

Average stunting

Average Wasting

Average Severe Wasting

Average Underweight

Total Country

Maximum Wasting

Maximum Severe Wasting

Total calculated fields: 5+

7.3 Number of Visualizations

Our report is visually rich and well-balanced with a mix of chart types for better understanding.

Visualization Type	Count	Examples
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Line Chart	1	Trend of indicators over years
Bar/Column Charts	5	Top/Bottom countries
Pie/Donut Charts	2	Malnutrition distribution
KPI Cards	2	Avg Overweight, Total Surveys
Funnel Chart	1	Funnel Chart Showing Sum of under 5 Population
Map Chart	1	Global malnutrition spread
Table	1	Outlier countries

- **Total visualizations used: 13**

All visuals are clean, not overloaded, and responsive to filters.

8. Conclusion / Observations

The Power BI project provided a deep and interactive view into the global malnutrition crisis. By visualizing key indicators such as **stunting**, **wasting**, **underweight**, **overweight**, and **severe wasting**, we were able to understand how malnutrition has impacted different countries over the years.

Key Observations:

- **Stunting** emerged as the most widespread issue, accounting for nearly **49%** of the total malnutrition cases in our dataset.
- Countries like **Yemen, Ethiopia, and Burundi** showed critically high levels of underweight and wasting, indicating regions of serious nutritional concern.
- **Vietnam** had the **highest under-5 population**, making it a key focus area for public health initiatives.
- Some developed countries such as the **USA and Uruguay** reflected rising **overweight percentages**, showing the **dual burden of malnutrition** — both undernutrition and overnutrition.
- The use of **interactive visuals** like KPIs, trend lines, pie charts, and maps helped us discover patterns quickly and clearly.

Final Insight:

This project proves that when data is visualized effectively, it not only tells a story but also supports better decision-making. Power BI helped us turn raw numbers into valuable insights that can guide public health actions.

9. Future Scope

While the current dashboard offers valuable insights into global malnutrition trends, there is strong potential to expand this project further in the future. Enhancements can make the report more dynamic, insightful, and impactful for researchers, policymakers, and health organizations.

Potential Future Improvements:

- **Time Series Forecasting:**
Integrate forecasting models to predict future trends in stunting, underweight, or overweight rates using Power BI's built-in analytics or machine learning tools.
- **Regional & Continent Grouping:**
Group countries into regions like **South Asia**, **Sub-Saharan Africa**, or **WHO zones** to understand malnutrition trends by geography.
- **Real-Time Data Integration:**
Connect the report with **live APIs** from sources like UNICEF or WHO for automatic data updates, keeping the dashboard current and actionable.
- **Nutrient-Level Analysis:**
Expand the dataset to include **micronutrient deficiencies** (like Vitamin A, Iron, and Iodine), offering a more complete picture of nutritional health.
- **Severity Scoring System:**
Create a "**Malnutrition Risk Score**" using weighted averages of multiple indicators to classify countries into low, medium, and high-risk zones.
- **Mobile or Web Deployment:**
Publish the report online or embed it into **public health dashboards** or websites for easier access and advocacy.
- **User-Specific Filters:**
Add slicers for gender, age groups, or economic classifications (if data available) to make the report more flexible and user-focused.

In Summary:

The project has laid a strong foundation, and with these improvements, it can become a powerful tool for global nutrition monitoring, awareness, and policymaking.

10. Appendix:

10.1 Source Code-

Dataset Link:

 <https://www.kaggle.com/datasets/ruchi798/malnutrition-across-the-globe>

10.2. GitHub & Project Demo Link

- GitHub Link: [Chaitanya GitHub Link](#)
- Demo Link: [Chaitanya Jadhav Demo Link](#)